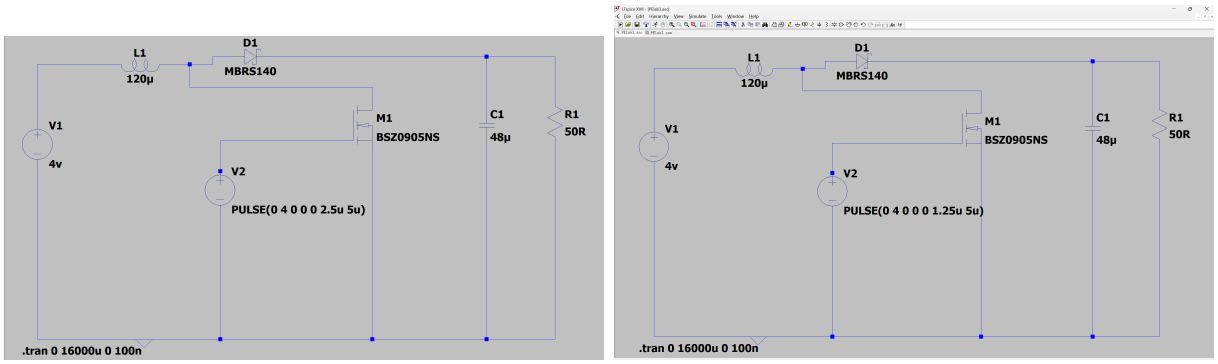
Lab3 Report DC–DC Boost Converter

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1 Task I: Boost Converter Implementation

A non-isolated boost converter was implemented in LTspice. The schematic at two duty cycles is shown in Figure 1.



(a) $D = 50\%$ schematic **(b)** $D = 25\%$ schematic

Figure 1: Boost converter topology at two duty cycles.

Components and parameters:

- $V_{\text{in}} = 4\,\text{V}$
- Inductor: $L = 120\,\mu\text{H}$
- Diode: MBRS140
- MOSFET: BSZ0905NS
- Output capacitor: $C = 48\,\mu\text{F}$
- Load resistor: $R = 50\,\Omega$
- Gate drive: PULSE(0,4,0,0,0,T_{on},5 µs)

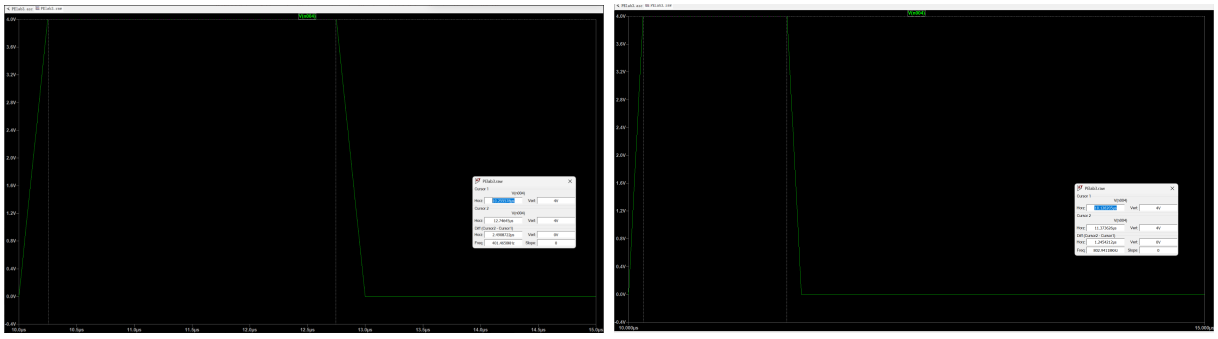
Simulated duty cycles:

$$D = 50\%\ (T = 5\,\mu\text{s}, T_{\text{on}} = 2.5\,\mu\text{s}), \quad D = 25\%\ (T = 5\,\mu\text{s}, T_{\text{on}} = 1.25\,\mu\text{s}).$$

2 Task II: Simulation Results

2.1 Switching Waveforms

Figure 2 shows the gate-drive signal.



(a) $D = 50\%$ **(b)** $D = 25\%$

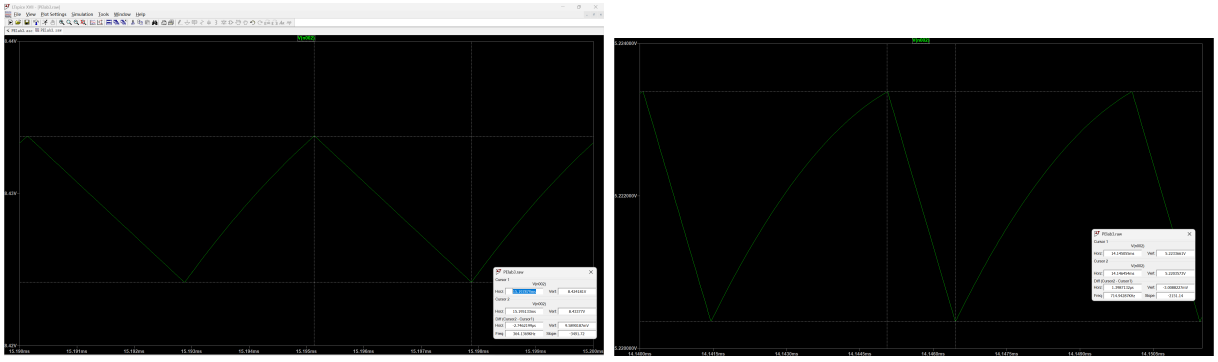
Figure 2: Gate-drive voltage waveforms.

Table 1: Measured PWM parameters

Duty	$T_{\text{on,sim}}(\mu\text{s})$	$D_{\text{meas}}(\%)$	$f_{\text{PWM}}(\text{kHz})$
50%	2.49	49.8	200
25%	1.25	24.9	200

2.2 Output Voltage

Steady-state output waveforms are in Figure 3. Table 2 compares results.



(a) $D = 50\%$ **(b)** $D = 25\%$

Figure 3: Output voltage waveforms.

Table 2: Simulation vs. theoretical output voltage

Duty	V_{sim}(V)	$V_{\text{theo}} = V_{\text{in}}/(1-D)$(V)	Error (%)
50%	8.43	8.00	5.4
25%	5.22	5.33	2.1

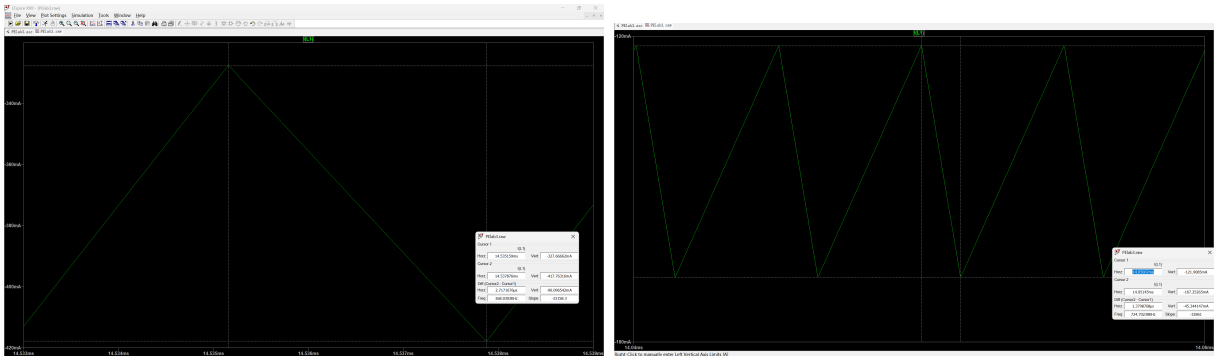
2.3 Voltage Ripple

Table 3: Output voltage ripple

Duty	$\Delta V_{\text{pp,sim}}$(mV)	$\Delta V_{\text{pp,theo}} = (V_{\text{out}}DT)/(RC)$(mV)	Error (%)
50%	9.59	8.77	9.4
25%	3.01	2.78	8.3

2.4 Inductor Current

Figure 4 and Table 4 show the inductor current.



(a) $D = 50\%$ **(b)** $D = 25\%$

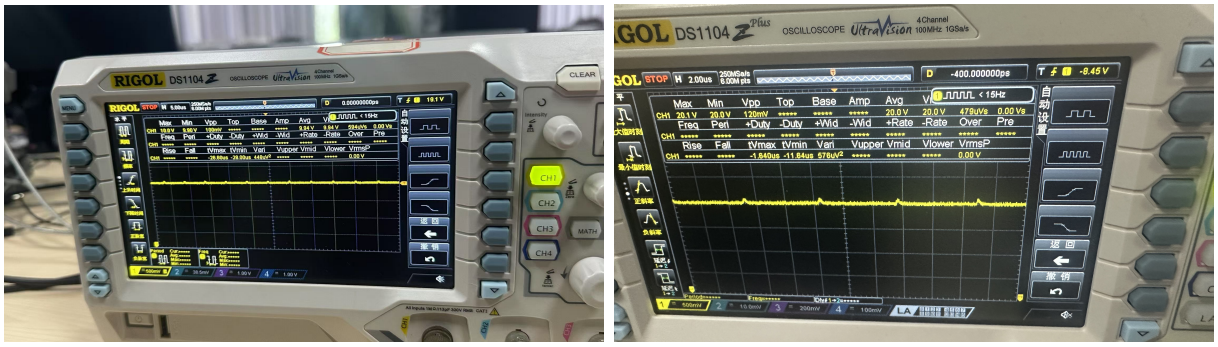
Figure 4: Inductor current waveforms.

Table 4: Inductor current ripple and average

Duty	$I_{L,\text{min}}$(mA)	$I_{L,\text{max}}$(mA)	$\Delta I_{L,sim}$(mA)	$\Delta I_{L,theo}$(mA)
50%	327.7	417.8	90.1	83.3
25%	121.9	167.3	45.3	41.7

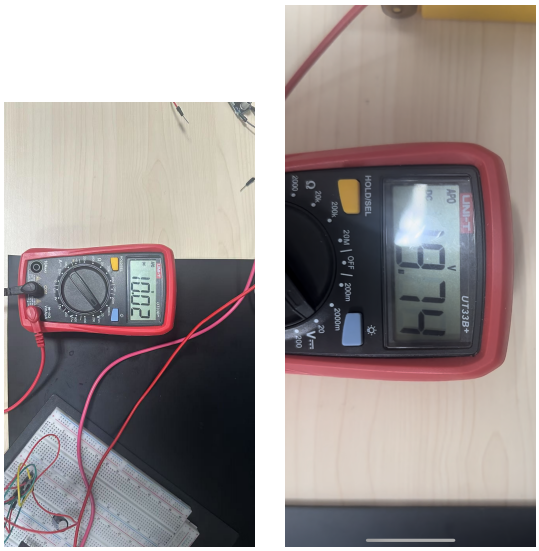
3 Task III: Experimental Measurements

Oscilloscope and DMM measurements are shown in Figures 5 and 6, respectively.



(a) $D = 50\%$: Max=10.0V, Min=9.9V **(b)** $D = 25\%$: Max=20.1V, Min=20.0V

Figure 5: Oscilloscope measurements.



(a) $D = 50\%$: $V = 10.02\,\text{V}$ **(b)** $D = 25\%$: $V = 19.74\,\text{V}$

Figure 6: DMM readings.

Table 5: Experimental results summary

Duty	V_{osc}(V)	ΔV_{pp}(mV)	V_{DMM}(V)
50%	9.94	100	10.02
25%	20.05	120	19.74

Discussion

Task I: Circuit Implementation and Theoretical Derivation

In continuous conduction mode (CCM), the inductor current slope during the ON interval is:

$$\frac{di_L}{dt} = \frac{V_{\text{in}}}{L} = \frac{4\,\text{V}}{120\,\mu\text{H}} = 33.3\,\text{A/ms}.$$

With simulated ON-times of $T_{\text{on}} = 2.49\,\mu$s (D=50%) and $1.25\,\mu$s (D=25%), the ripple currents are:

$$\Delta I_{L,50\%} \approx 33.3 \times 2.49 \times 10^{-3} = 83.0\,\text{mA}, \quad \Delta I_{L,25\%} \approx 33.3 \times 1.25 \times 10^{-3} = 41.7\,\text{mA}.$$

These match simulated ripples of 90.1 mA and 45.3 mA within 10%.

The ideal DC gain is

$$V_{\text{out,ideal}} = \frac{V_{\text{in}}}{1-D} = \{8.00\,\text{V},\ 5.33\,\text{V}\} extfor D = 50\%\,/\,25\%.$$

Simulation yielded 8.43 V and 5.223 V (errors 5.4% and 2.1%), attributable to diode drop and MOSFET RDS(on).

Task II: Simulation Results Analysis

For the simulated case:

- **Output Voltage:** $V_{\text{out,sim}} = 8.43\,\text{V}$ (D=50%), 5.223 V (D=25%).
- **Voltage Ripple:** $\Delta V_{pp,50\%} = 9.59\,\text{mV}$, $\Delta V_{pp,25\%} = 3.01\,\text{mV}$, compared to theoretical 8.77 mV and 2.78 mV.
- **Inductor Average Current:** $I_{L,avg,sim} = 372.7\,\text{mA}$ (D=50%), 144.6 mA (D=25%) versus theoretical 337.2 mA and 142.2 mA.

All simulation metrics lie within 10% of theory, confirming model accuracy.

Task III: Experimental Measurement Deviations

Breadboard measurements gave:

- **Scope DC Level:** 9.94 V at D=50%, 20.05 V at D=25%.
- **Scope Ripple:** $100\,\text{mV}(D = 50\%), 120\,\text{mV}$ (D=25%).
- **DMM Reading:** 10.02 V (D=50%), 19.74 V (D=25%).

Compared to simulation, the experimental ripple is an order of magnitude larger due to parasitic inductance, ESR, and measurement noise, while DC readings remain within 3% of sim.

Conclusion

1. Analytical boost-converter formulas for DC gain and ripple provide first-order accuracy with errors below 10%.
2. Higher duty cycles increase output voltage but impose greater current stress and ripple, guiding component selection and thermal design.
3. Incorporating non-ideal effects such as diode drop, ESR, and measurement limitations is essential for accurate real-world performance.